

Flow Separation Criterion

Continuity Equation (curvilinear)

$$\frac{1}{1+Kn} \frac{\partial(\ell u_s)}{\partial s} + \frac{\partial(\ell u_n)}{\partial n} + \frac{K}{1+Kn} (\ell u_n) = 0$$

At wall, no slip and no permeability.

$$(u_s, u_n) \Big|_{n=0, \text{wall}} = (0, 0)$$

→ At $n=0$, wall, s = any value in the domain

$$u_s = 0 \text{ At all values of } s$$

$$u_n = 0$$

$$\text{Therefore, } \frac{\partial u_s}{\partial s} = 0, \frac{\partial u_n}{\partial s} = 0$$

$$\rightarrow \frac{1}{1+K(0)} \frac{\partial(\ell u_s)}{\partial s} + \frac{\partial(\ell u_n)}{\partial n} + \frac{K}{1+K(0)} (\ell u_n) = 0, \text{ at wall}$$

$$\frac{\partial(\ell u_s)}{\partial s} + \frac{\partial(\ell u_n)}{\partial n} = 0$$

$$\ell \frac{\partial u_s}{\partial s} + u_s \frac{\partial \ell}{\partial s} + \ell \frac{\partial u_n}{\partial n} + u_n \frac{\partial \ell}{\partial n} = 0$$

$$\therefore \ell \frac{\partial u_n}{\partial n} = 0; \ell > 0$$

Therefore, $\frac{\partial u_n}{\partial n} \Big|_{n=0} = 0$ at wall for all s

Flow Separation Criterion

Streamwise/tangential momentum equation (curvilinear)

$$\rho \left(\frac{u_s}{1+Kn} \frac{\partial u_s}{\partial s} + \frac{\partial u_s}{\partial n} + \frac{K u_s u_n}{1+Kn} \right) = - \frac{1}{1+Kn} \frac{\partial P}{\partial s} + \frac{1}{1+Kn} \frac{\partial \tau_{ss}}{\partial s}$$

$$0 = - \frac{\partial P}{\partial s} + \frac{\partial \tau_{ss}}{\partial s} + \frac{\partial \tau_{sn}}{\partial n} + 2K \tau_{sn} \text{ at wall} + \frac{\partial \tau_{sn}}{\partial n} + \frac{2K}{1+Kn} \tau_{sn}$$

$$\tau_{ss} = \frac{4}{3} \mu \left(\frac{1}{1+Kn} \frac{\partial u_s}{\partial s} + \frac{K u_n}{1+Kn} \right) - \frac{2}{3} \mu \frac{\partial u_n}{\partial n} \text{ at wall } (n=0, u_s=0, u_n=0)$$

$$\frac{\partial \tau_{ss}}{\partial s} = \frac{4}{3} \mu \left(\frac{1}{1+Kn} \frac{\partial^2 u_s}{\partial s^2} + \frac{\partial u_s}{\partial s} \frac{\partial}{\partial s} \left(\frac{1}{1+Kn} \right) + \frac{\partial u_n}{\partial s} \left(\frac{K}{1+Kn} \right) + \frac{\partial}{\partial s} \left(\frac{K}{1+Kn} \right) \right)$$

$$u_s(s) \Big|_{n=0} = 0 \rightarrow \frac{\partial u_s(s)}{\partial s} \Big|_{n=0} = 0 \rightarrow \frac{\partial^2 u_s(s)}{\partial s^2} \Big|_{n=0} = 0 - \frac{2}{3} \mu \frac{\partial}{\partial s} \left(\frac{\partial u_n}{\partial n} \right) \Big|_{n=0}$$

$$u_n(s) \Big|_{n=0} = 0 \rightarrow \frac{\partial u_n(s)}{\partial s} \Big|_{n=0} = 0 \left[\frac{\partial \tau_{ss}}{\partial s} \right] = - \frac{2}{3} \mu \frac{\partial}{\partial s} \left(\frac{\partial u_n}{\partial n} \right) \Big|_{n=0}$$

$$\frac{\partial u_n(s)}{\partial n} \Big|_{n=0} = 0 \rightarrow \frac{\partial}{\partial s} \left(\frac{\partial u_n}{\partial n} \right) \Big|_{n=0} = 0$$

$$\therefore \frac{\partial \tau_{ss}}{\partial s} = - \frac{2}{3} \mu(0) \rightarrow \frac{\partial \tau_{ss}}{\partial s} = 0$$

$$\therefore \left(0 = - \frac{\partial P}{\partial s} + \frac{\partial \tau_{sn}}{\partial n} + 2K \tau_{sn} \right) \Big|_{n=0, \text{ wall}}$$

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$$\tau_{sn} = \mu \left(\frac{\partial u_s}{\partial n} - \frac{K u_s}{1+Kn} + \frac{1}{1+Kn} \frac{\partial u_n}{\partial s} \right)$$

$u_s=0$ $\frac{\partial u_n}{\partial s}=0$

$$\tau_{sn} = \mu \frac{\partial u_s}{\partial n} \Big|_{n=0} \text{ at wall because}$$

$$\therefore \left(0 = -\frac{\partial p}{\partial s} + \frac{\partial \tau_{sn}}{\partial n} + 2K\mu \frac{\partial u_s}{\partial n} \right) \Big|_{n=0} \quad \begin{matrix} u_s(s)=0 \text{ and } u_n(s)=0 \\ n=0 \end{matrix} \quad \text{and } \frac{\partial u_n(s)}{\partial s} \Big|_{n=0} = 0$$

$$\frac{\partial \tau_{sn}}{\partial n} \Big|_{n=0} = \mu \left(\frac{\partial^2 u_s}{\partial n^2} - \frac{K}{1+Kn} \frac{\partial u_s}{\partial n} - \frac{K}{1+Kn} \frac{\partial}{\partial n} \left(\frac{1}{1+Kn} \right) + \frac{1}{1+Kn} \frac{\partial^2 u_n}{\partial n \partial s} \right) \Big|_{n=0}$$

$$\therefore \frac{\partial \tau_{sn}}{\partial n} \Big|_{n=0} = \mu \left(\frac{\partial^2 u_s}{\partial n^2} - \frac{K}{1+K(0)} \frac{\partial u_s}{\partial n} + \frac{1}{1+K(0)} \frac{\partial^2 u_n}{\partial n \partial s} \right) \Big|_{n=0} + \frac{\partial u_n}{\partial s} \Big|_{n=0} \frac{\partial}{\partial n} \left(\frac{1}{1+Kn} \right) \Big|_{n=0}$$

$$\frac{\partial^2 u_n}{\partial n \partial s} = \frac{\partial}{\partial s} \left(\frac{\partial u_n}{\partial n} \right); \quad \frac{\partial^2 u_n}{\partial n \partial s} (s, n=0) = \frac{\partial}{\partial s} \left(\frac{\partial u_n}{\partial n} \Big|_{n=0} \right) = 0$$

$$\frac{\partial u_n(s)}{\partial n} \Big|_{n=0} = 0 \rightarrow \frac{\partial}{\partial s} \left(\frac{\partial u_n(s)}{\partial n} \Big|_{n=0} \right) = 0$$

$$\rightarrow \left(\frac{\partial \tau_{sn}}{\partial n} = \mu \left(\frac{\partial^2 u_s}{\partial n^2} - K \frac{\partial u_s}{\partial n} \right) \right) \Big|_{n=0}$$

$$\frac{\partial \tau_{sn}}{\partial n} \Big|_{n=0} = \mu \frac{\partial^2 u_s}{\partial n^2} \Big|_{n=0} - K\mu \frac{\partial u_s}{\partial n} \Big|_{n=0}$$

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$$\therefore 0 = -\left.\frac{\partial p}{\partial s}\right|_{n=0} + \left.\frac{\partial^2 u_s}{\partial n}\right|_{n=0} + 2K\mu \left.\frac{\partial u_s}{\partial n}\right|_{n=0}$$

$$\therefore 0 = -\left.\frac{\partial p}{\partial s}\right|_{n=0} + \mu \left.\frac{\partial^2 u_s}{\partial n^2}\right|_{n=0} - K\mu \left.\frac{\partial u_s}{\partial n}\right|_{n=0} + 2K\mu \left.\frac{\partial u_s}{\partial n}\right|_{n=0}$$

$$\therefore 0 = -\left.\frac{\partial p}{\partial s}\right|_{n=0} + \mu \left.\frac{\partial^2 u_s}{\partial n^2}\right|_{n=0} + K\mu \left.\frac{\partial u_s}{\partial n}\right|_{n=0}$$

$$\boxed{\left.\frac{\partial p}{\partial s}\right|_{n=0} = \mu \left.\frac{\partial^2 u_s}{\partial n^2}\right|_{n=0} + K\mu \left.\frac{\partial u_s}{\partial n}\right|_{n=0}}$$

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Flow Separation Criterion

$$\left. \frac{\partial p}{\partial s} \right|_{n=0} = \mu \left. \frac{\partial^2 u_s}{\partial n^2} \right|_{n=0} + K\mu \left. \frac{\partial u_s}{\partial n} \right|_{n=0} \quad (1)$$

Thin Boundary Layer Approximation

Assume boundary layer before flow separation is thin. $\delta \ll 1$

$$O(\delta^2) \approx 0$$

Use Taylor series approximation to $\frac{\partial u_s}{\partial n}$

$$\left. \frac{\partial u_s}{\partial n} \right|_{n=\delta} = \left. \frac{\partial u_s}{\partial n} \right|_{n=0} + \delta \left. \frac{\partial^2 u_s}{\partial n^2} \right|_{n=0} + \cancel{O(\delta^2)}$$

$\left. \frac{\partial u_s}{\partial n} \right|_{n=\delta} \approx 0$, because by definition $n=\delta$ is at the boundary layer edge when the streamwise/tangential velocity has recovered 99% of its inviscid approximation of streamwise/tangential velocity and reaches an asymptote at the inviscid approximation of velocity (streamwise/tangential)

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Therefore, normal gradient of streamwise/tangential velocity is approximately 0 at the edge of the boundary layer at $n=\delta$

$$\therefore \left. \frac{\partial u_s}{\partial n} \right|_{n=\delta} = 0$$

$$0 = \left. \frac{\partial u_s}{\partial n} \right|_{n=0} + \delta \left. \frac{\partial^2 u_s}{\partial n^2} \right|_{n=0}$$

$$-\frac{1}{\delta} \left. \frac{\partial u_s}{\partial n} \right|_{n=0} = \left. \frac{\partial^2 u_s}{\partial n^2} \right|_{n=0}$$

$$\text{Since } \left. \frac{\partial p}{\partial s} \right|_{n=0} = \mu \left. \frac{\partial^2 u_s}{\partial n^2} \right|_{n=0} + K\mu \left. \frac{\partial u_s}{\partial n} \right|_{n=0},$$

$$\therefore \left. \frac{\partial p}{\partial s} \right|_{n=0} = \mu \left(-\frac{1}{\delta} \left. \frac{\partial u_s}{\partial n} \right|_{n=0} \right) + K\mu \left. \frac{\partial u_s}{\partial n} \right|_{n=0}$$

and $K = \frac{1}{R}$ where K is curvature and R is radius of curvature

$$\therefore \left. \frac{\partial p}{\partial s} \right|_{n=0} = -\frac{\mu}{\delta} \left. \frac{\partial u_s}{\partial n} \right|_{n=0} + \frac{\mu}{R} \left. \frac{\partial u_s}{\partial n} \right|_{n=0}$$

$$\therefore \left. \frac{\partial p}{\partial s} \right|_{n=0} = \left(-\frac{1}{\delta} + \frac{1}{R} \right) \mu \left. \frac{\partial u_s}{\partial n} \right|_{n=0}$$

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Since,

$$\left. \frac{\partial P}{\partial s} \right|_{n=0} = \left(-\frac{1}{\delta} + \frac{1}{R} \right) \mu \left. \frac{\partial u_s}{\partial n} \right|_{n=0}$$

$$\left. \frac{\partial P_s}{\partial s} \right|_{n=0} = \left(\frac{\delta - R}{R\delta} \right) \mu \left. \frac{\partial u_s}{\partial n} \right|_{n=0}$$

$$\frac{R\delta}{\delta - R} \left. \frac{\partial P}{\partial s} \right|_{n=0} = \mu \left. \frac{\partial u_s}{\partial n} \right|_{n=0}$$

or

$$-\frac{R\delta}{R - \delta} \left. \frac{\partial P}{\partial s} \right|_{n=0} = \mu \left. \frac{\partial u_s}{\partial n} \right|_{n=0} \quad \text{since typically } R > \delta$$

or

$$\frac{R\delta}{\delta - R} \left. \frac{\partial P}{\partial s} \right|_{n=0} = \tau_{\text{wall}}$$

or

$$-\frac{R\delta}{R - \delta} \left. \frac{\partial P}{\partial s} \right|_{n=0} = \tau_{\text{wall}} \quad \text{since typically } R > \delta$$

Flow Separation Criterion

For flow to separate, shear stress that the flow exerts on the wall must become 0 and reverse signs and become negative

$$\tau_{wall} \leq 0$$

Since

$$\left. \frac{R\delta}{\delta-R} \frac{\partial p}{\partial s} \right|_{n=0} = \tau_{wall}$$

or

$$-\left. \frac{R\delta}{R-\delta} \frac{\partial p}{\partial s} \right|_{n=0} = \tau_{wall}$$

For flow to separate

$$\left. \frac{R\delta}{\delta-R} \frac{\partial p}{\partial s} \right|_{n=0} \leq 0$$

or

$$-\left. \frac{R\delta}{R-\delta} \frac{\partial p}{\partial s} \right|_{n=0} \leq 0 \Rightarrow \left. \frac{R\delta}{R-\delta} \frac{\partial p}{\partial s} \right|_{n=0} \geq 0$$